Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The area and perimeter of a rectangle are A and P respectively. Then P and A satisfy the inequality.

(A) $P + A > PA$

(B) $P^2 \leq A$

(C) $A - P < 2$

(D) $P^2 \geq 16A$

Q.2

If $\left|\frac{12x}{4x^2+9}\right| \geq 1$ for all real values of x, the inequality being satisfied only if $|x|$ is equal to

(A) $\frac{3}{2}$

(B) $\frac{2}{3}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

Q.3

A manufacturer has 600 litres of a 12% solution of acid. How many litres of a 30% acid solution must be added to it so that acid content in the resulting mixture will be more than 15% but less than 18% ?

(A) more than 120 litres but less than 300 litres

(B) more than 140 litres but less than 600 litres

(C) more than 100 litres but less than 280 litres

(D) more than 160 litres but less than 500 litres

Q.4

The number of integral roots of the equation $|x-1| + |x+2| - |x-3| = 4$ is

(A) 0

(B) 1

(C) 2

(D) 4

@IITJEE_Advanced

Q.5

A man wants to cut three lengths from a single piece of board of length $91\ \text{cm}$. The second length is to be $3\ \text{cm}$ longer than the shortest and the third length is to be twice as long as the shortest. The possible length of the shortest board, if the third piece is to be at least $5\ \text{cm}$ longer than the second, is

(A) less than $8\ \text{cm}$

(B) greater than or equal to $8\ \text{cm}$ but less than or equal to $22\ \text{cm}$

(C) less than $22\ \text{cm}$

(D) greater than $22\ \text{cm}$

Q.6

If a, b, c are distinct positive real numbers then the expression $(\text{b} + \text{c} - \text{a})(\text{c} + \text{a} - \text{b})(\text{a} + \text{b} - \text{c}) - \text{abc}$ is

(A) positive

(B) negative

(C) non-positive

(D) non-negative

Q.7

The set of real values of x satisfying $|x - 1| \leq 3$ and $|\text{x} - 1| \geq 1$ is

(A) $[2, 4]$

(B) $(-\infty, 2] \cup [4, +\infty)$

(C) $[-2, 0] \cup [2, 4]$

(D) None of these

Q.8

Solution set of equation $\left|1 - \log_{\frac{1}{6}} x\right| + \left|\log_{\frac{1}{2}} x\right| + 2 = \left|3 - \log_{\frac{1}{6}} x + \log_{\frac{1}{2}} x\right|$ is $\left[\frac{a}{b}, a\right], a, b \in N$, then the value of $a + b$ is

(A) 5

(B) 6

(C) 7

(D) 8

Q.9

@IITJEE_Advanced

Set of values of x satisfying the inequality $\frac{x^2+6x-7}{|x+4|} < 0$ is /are

(A) $(-\infty, -7)$

(B) $(-7, 4)$

(C) $(-7, -4) \cup (-4, 1)$

(D) $(1, \infty)$

Q.10

A vertex of a feasible region by the linear constraints $3x + 4y \leq 18, 2x + 3y \geq 3$ and $x, y \geq 0$, is

(A) $(0, 2)$

(B) $(4.8, 0)$

(C) $(0, 3)$

(D) None of these

Q.11

For $\mathrm{x} \in \mathrm{R}$, $\langle \mathrm{x} \rangle$ is defined as follows:

$$\langle x \rangle = \begin{cases} x + 1, & 0 \leq x < 2 \\ |x - 4|, & x \geq 2 \end{cases}$$

Then the solution set of the equation $\langle \mathrm{x} \rangle^2 + \mathrm{x} = \langle \mathrm{x} \rangle + \mathrm{x}^2$ is

(A) $\{-1, 1\}$

(B) $[2, \infty)$

(C) $[0, 2)$

(D) $\{0, 2\}$

Q.12

If $\frac{|x+3|+x}{x+2} > 1$, then $x \in$

(A) $(-5, -2)$

(B) $(-1, \infty)$

(C) $(-5, -2) \cup (-1, \infty)$

(D) None of these

Q.13

The equation $||x - 1| + a| = 4$ can have real solutions for x if ' a ' belongs to the interval

(A) $(-\infty, +\infty)$

@IITJEE_Advanced

(B) $(-\infty, 4]$

(C) $(4, +\infty)$

(D) $[-4, 4]$

Q.14

If $\log_{245} 175 = a, \quad \log_{1715} 875 = b$, then $\frac{1 - ab}{a - b} =$

(A) 5

(B) 6

(C) 2

(D) 1

Q.15

The graph of inequations $x \leq y$ and $y \leq x + 3$ is located in

(A) II quadrant

(B) I, II quadrants

(C) I, II, III quadrants

(D) II, III, IV quadrants

Q.16

Consider the following statements.

I. Solution set of the inequality $-15 < \frac{3(x-2)}{5} \leq 0$ is $(-23, 2]$

II. Solution set of the inequality

$$7 \leq \frac{3x + 11}{2} \leq 11 \text{ is } \left[1, \frac{11}{3}\right]$$

III. Solution set of the inequality

$$-5 \leq \frac{2 - 3x}{4} \leq 9 \text{ is } [-1, 1] \cup [3, 5]$$

Choose the correct option

(A) Only I and II are true.

(B) Only II and III are true.

(C) Only I and III are true.

(D) All are true.

@IITJEE_Advanced

Q.17

The marks obtained by a student of class XI in first and second terminal examinations are 62 and 48 , respectively. The minimum marks he should get in the annual examination to have an average of at least 60 marks, are

(A) 70

(B) 50

(C) 74

(D) 48

Q.18

Solution set of the inequality $\frac{1}{2^x-1} > \frac{1}{1-2^{x-1}}$ is

(A) $(1, \infty)$

(B) $(0, \log_2(4/3))$

(C) $(-1, \infty)$

(D) $(0, \log_2(4/3)) \cup (1, \infty)$

Q.19

If the equation $2^x + 4^y = 2^y + 4^x$ is solved for y in terms of x where $x < 0$, then the sum of the solutions is

(A) $x \log_2 (1 - 2^x)$

(B) $x + \log_2 (1 - 2^x)$

(C) $\log_2 (1 - 2^x)$

(D) $x \log_2 (2^x + 1)$

Q.20

A company manufactures cassettes. Its cost and revenue functions are $\mathrm{C(x)} = 26000 + 30\mathrm{x}$ and $\mathrm{R(x)} = 43\mathrm{x}$, respectively, where x is the number of cassettes produced and sold in a week. The number of cassettes must be sold by the company to realise some profit, is

(A) more than 2000

(B) less than 2000

(C) more than 1000

(D) less than 1000

@IITJEE_Advanced

Q.1 Answer:

$P^2 \geq 16A$

Solution:

Let x be the length and y be the breadth of a rectangle. $A = xy, P = 2(x+y)$ **Since** $(x+y)^2 \geq 4xy$

$\Rightarrow \left(\frac{P}{2}\right)^2 \geq 4A \Rightarrow P^2 \geq 16A$

Q.2 Answer:

$\frac{3}{2}$

Solution:

$\left|\frac{12x}{4x^2+9}\right| \geq 1 \Rightarrow \frac{12|x|}{4x^2+9} \geq 1$

$\because 4x^2 + 9 > 0$

$\Rightarrow 4x^2 - 12|x| + 9 \leq 0$

$\Rightarrow 4|x|^2 - 12|x| + 9 \leq 0$

$\Rightarrow (2|x| - 3)^2 = 0 \Rightarrow |x| = \frac{3}{2}$

Q.3 Answer:

more than 120 litres but less than 300 litres

Solution:

Let x litres of 30% acid solution is required to be added. Then, Total mixture $= (x + 600)$ litres

$\therefore$ 30% **of** $x + 12\%$ **of** $600 > 15\%$ **of** $(x + 600)$ **and** 30% **of** $x + 12\%$ **of** $600 < 18\%$ **of** $(x + 600)$

or $\frac{30x}{100} + \frac{12}{100}(600) > \frac{15}{100}(x + 600)$

and $\frac{30x}{100} + \frac{12}{100}(600) < \frac{18}{100}(x + 600)$

or $30x + 7200 > 15x + 9000$ **and** $30x + 7200 < 18x + 10800$

or $15x > 1800$ **and** $12x < 3600$ **or** $x > 120$ **and** $x < 300$ **i.e.** $120 < x < 300$

Thus, the number of litres of the 30% solution of acid will have to be more than 120 litres but less than 300 litres

Q.4 Answer:

2

Solution:

$|x-1| + |x+2| - |x-3| = 4$**, has three absolute value expressions, thus we divide the problem into four intervals :**

(i) If $x < -2$ then

$-(x-1) - (x+2) + (x-3) = 4 \Rightarrow x = -8$

(ii) If $-2 \leq x < 1$, then

@IITJEE_Advanced

$-(x-1)+(x+2)+(x-3)=4$

$\Rightarrow \mathrm{x}=4 \notin[-2,1)$, **hence rejected**

(iii) If $1 \leq x<3$, **then** $(x-1)+(x+2)+(x-3)=4 \Rightarrow x=2$

(iv) If $x \geq 3$, **then**

$(\mathrm{x}-1)+(\mathrm{x}+2)-(\mathrm{x}-3)=4 \Rightarrow \mathrm{x}=0 \notin[3, \infty)$

hence rejected

∴ Solution set is $\{-8,2\}$ **and both are integers**

Q.5 Answer:

greater than or equal to $8 \mathrm{~cm}$ but less than or equal to $22 \mathrm{~cm}$

Solution:

Let the shortest side be $\times \mathrm{cm}$. **Then, by given condition, second length** $=x+3 \mathrm{~cm}$

Third length $=2 \times \mathrm{cm}$

Also given, total length $=91$ **Hence, sum of all the three lengths should be less than or equal to 91**

$\Rightarrow x+x+3+2x \leq 91 \Rightarrow 4x+3 \leq 91$

Subtracting (-3) **to each term,** $-3+4x+3 \leq 91-3 \Rightarrow 4x \leq 88 \Rightarrow \frac{4x}{4} \leq \frac{88}{4} \Rightarrow x \leq \frac{88}{4}$

$\Rightarrow \quad x \leq 22 \mathrm{~cm} \quad \ldots$

Again, given that Third length $\geq$ **Second length** $+5 \Rightarrow \quad 2x \geq (x+3)+5 \Rightarrow 2x \geq x+(3+5)$

Transferring the term x **to L.H.S.,** $2\mathrm{x}-\mathrm{x} \geq 8 \Rightarrow \mathrm{x} \geq 8 \quad \ldots$

From equations (i) and (ii), length of shortest board should be greater than or equal to 8 but less than or equal to 22, i.e., $8 \leq x \leq 22$

Q.6 Answer:

negative

Solution:

$(b+c-a)(c+a-b)(a+b-c)-abc$

Without loss of generality we can assume, $\mathrm{a}>\mathrm{b}>\mathrm{c}$

Applying A.M, G. M. pair wise

$(\mathrm{b}+\mathrm{c})>2\sqrt{\mathrm{bc}}$

$(a+c)>2\sqrt{ac}$

$(a+b)>2\sqrt{(ac)}$

Multiplying equation (i), (ii), (iii), we get $(a+b)(b+c)(c+a)>8abc$

Let us put $\mathrm{b}+\mathrm{c}=2\mathrm{p}; \mathrm{c}+\mathrm{a}=2\mathrm{q}; \mathrm{a}+\mathrm{b}=2\mathrm{r}$

$\Rightarrow \mathrm{a}=-\mathrm{p}+\mathrm{r}+\mathrm{q}; \mathrm{b}=\mathrm{p}-\mathrm{q}+\mathrm{r}; \mathrm{c}=\mathrm{p}+\mathrm{q}-\mathrm{r}$

$\Rightarrow 2p.2q \cdot 2r \geq r \geq 8(q+r+p);(p+r-q)$

$(p+q-r)$

$\Rightarrow pqr \geq (q+r-p)(p+r-q)(p+r-r)$

Replacing $\mathrm{p}, \mathrm{q}, \mathrm{r}$ **by a** bc

@IITJEE_Advanced

Without loss of generality, we get $abc \geq (a+b-c)(b+c-a)(c+a-b)$

$\Rightarrow$ **The required expression is always negative**

Q.7 Answer:

$[-2,0] \cup [2,4]$

Solution:

$|x-1| \leq 3 \Rightarrow -3 \leq x-1 \leq 3 \Rightarrow -2 \leq x \leq 4$

and $|x-1| \geq 1 \Rightarrow x-1 \leq -1$ **or** $x-1 \geq 1$

$\Rightarrow x \leq 0$ **or** $x \geq 2$

Taking the common values of x, we get

$x \in [-2,0] \cup [2,4]$

Q.8 Answer:

7

Solution:

If $|a+b+c| = |a| + |b| + |c|$ **then** a, b, c **have same sign**

$\left|1 - \log_{1/6} x\right| + |-\log_2 x| + |2| = |3 - \log_{1/6} x - \log_2 x|$

$\therefore 1 - \log_{1/6} x \geqslant 0$

$\frac{1}{6} \leqslant x$

$-\log_2 x \geqslant 0$

$x \leqslant 1$

$\therefore x \in \left[\frac{1}{6}, 1\right]$, $a = 1$ **and** $b = 6$

$a + b = 7$

Q.9 Answer:

$(-7,-4) \cup (-4,1)$

Solution:

$\frac{x^2+6x-7}{|x+4|} < 0$

$\Rightarrow x^2 + 6x - 7 < 0$, **provided** $x + 4 \neq 0$

$[\because |x+4| > 0$ **if** $x \neq -4]$

$\Rightarrow (x+7)(x-1) < 0, x \neq -4 \Rightarrow -7 < x < 1$

$x \neq -4$

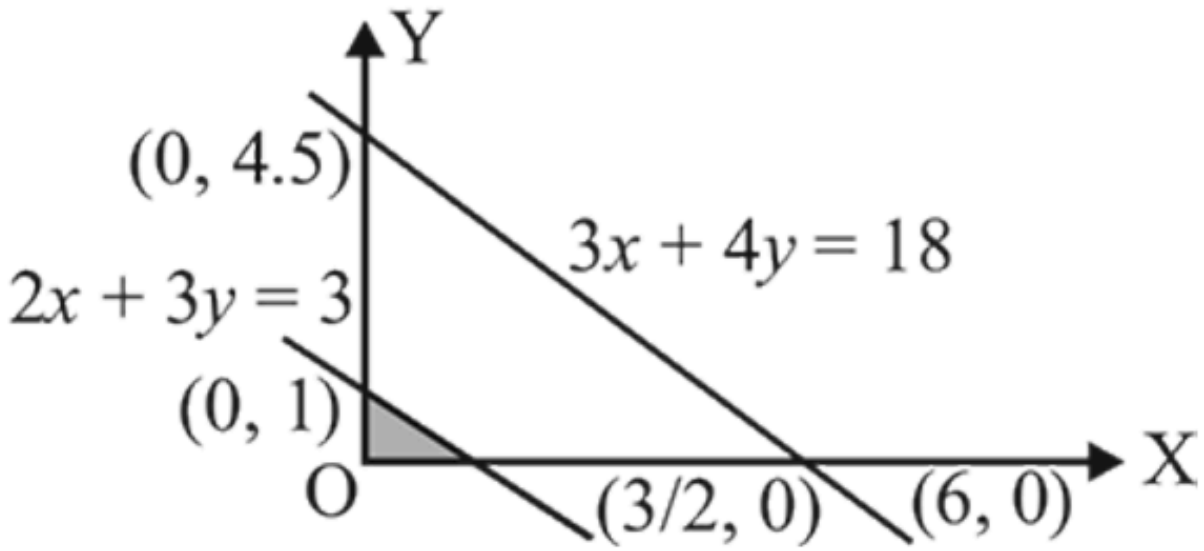
$\therefore \mathrm{x} \in (-7,-4) \cup (-4,1)$

Q.10 Answer:

None of these

Solution:

@IITJEE_Advanced



Q.11 Answer:

$\{0, 2\}$

Solution:

Case 1 : Let $0 \leq x < 2$, **then** $\langle x \rangle = (x + 1)$ **and the equation becomes** $(x + 1)^2 + x = (x + 1) + x^2$

$\Rightarrow 2x = 0 \Rightarrow x = 0$

Case 2 : Let $x \geq 2$ **then** $\langle x \rangle = |x - 4|$ **and the equation becomes** $|x - 4|^2 + x = |x - 4| + x^2$

$\Rightarrow x^2 - 8x + 16 + x = |x - 4| + x^2$

$\Rightarrow |x - 4| = 16 - 7x$

$\therefore x - 4 = \pm(16 - 7x)$, **provided** $16 - 7x \geq 0$

$\therefore x = \frac{5}{2}$ **or 2, but for** $x = \frac{5}{2}, 16 - 7x < 0$,

hence rejected $\therefore x = 2$. **The solution set is** $\{0, 2\}$

Q.12 Answer:

$(-5, -2) \cup (-1, \infty)$

Solution:

We have $\frac{|x+3|+x}{x+2} > 1$

$\Rightarrow \quad \frac{|x+3|+x}{x+2} - 1 > 0 \Rightarrow \frac{|x+3|-2}{x+2} > 0$

Now, two cases arise :

Case I : When $x + 3 \geq 0$, **i.e.** $x \geq -3$. **Then,** $\frac{|x+3|-2}{x+2} > 0 \Rightarrow \frac{x+3-2}{x+2} > 0$

$\Rightarrow \frac{x+1}{x+2} > 0$

$\Rightarrow \{(x + 1) > 0$ **and** $x + 2 > 0\}$

or $\{x + 1 < 0$ **and** $x + 2 < 0\}$

$\Rightarrow \{x > -1$ **and** $x > -2\}$ **or** $\{x < -1$ **and** $x < -2\}$

@IITJEE_Advanced

$\Rightarrow x > -1$ **or** $x < -2$

$\Rightarrow x \in (-1, \infty)$ **or** $x \in (-\infty, -2)$

$\Rightarrow x \in (-3, -2) \cup (-1, \infty)$ [**Since** $x \geq -3$] **. . . (i)**

Case II : When $x + 3 < 0$**, i.e.** $x < -3$ $\frac{|x+3|-2}{x+2} > 0 \Rightarrow \frac{-x-3-2}{x+2} > 0$

$\Rightarrow \frac{-(x+5)}{x+2} > 0 \Rightarrow \frac{x+5}{x+2} < 0$

$\Rightarrow (x + 5 < 0$ **and** $x + 2 > 0)$ **or** $(x + 5 > 0)$

and $x + 2 < 0) \Rightarrow (x < -5$ **and** $x > -2)$

or $(x > -5$ **and** $x < -2$ **) it is not possible.** $\Rightarrow x \in (-5, -2)$ **. . . (ii)**

Combining (i) and (ii), the required solution is $x \in (-5, -2) \cup (-1, \infty)$

Q.13 Answer:

$(-\infty, 4]$

Solution:

$||x - 1| + a| = 4 \Rightarrow |x - 1| + a = \pm 4$

$\Rightarrow |x - 1| = -a \pm 4$

The above equation holds if $-a + 4 \geq 0$ **or** $-a - 4 \geq 0$

$\Rightarrow a \leq 4$ **or** $a \leq -4 \Rightarrow a \in (-\infty, 4] \cup (-\infty, -4]$

$\Rightarrow a \in (-\infty, 4]$

Q.14 Answer:

5

Solution:

$175 = 5^2 . 7,\ 245 = 5.7^2,\ 875 = 5^3 . 7,\ 1715 = 5.7^3$

Let $\alpha = \log 5, \beta = \log 7$

$a = \frac{\log 175}{\log 245} = \frac{2\alpha + \beta}{\alpha + 2\beta}$

$b = \frac{\log 875}{\log 1715} = \frac{3\alpha + \beta}{\alpha + 3\beta}$

$\frac{1 - ab}{a - b} = \frac{\alpha + 2\beta\alpha + 3\beta - 2\alpha + \beta 3\alpha + \beta}{2\alpha + \beta\alpha + 3\beta - \alpha + 2\beta 3\alpha + \beta}$

$= \frac{5\beta^2 - \alpha^2}{\beta^2 - \alpha^2} = 5$

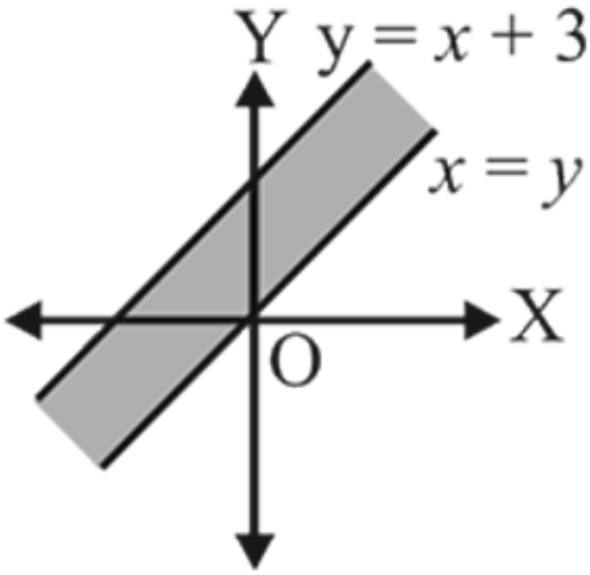
Q.15 Answer:

I, II, III quadrants

Solution:

@IITJEE_Advanced

The shaded area is the required area given in graph as below.



Hence, it is in I, II and III quadrant.

Q.16 Answer:

Only I and II are true.

Solution:

I. $-75 < 3x - 6 \Rightarrow -23 < x; \quad 3x - 6 \leq 0 \Rightarrow x \leq 2$

II. $14 \leq 3x + 11 \Rightarrow 3 \leq 3x \Rightarrow 1 \leq x$

$3x + 11 \leq 22 \Rightarrow 3x \leq 11 \Rightarrow x \leq \frac{11}{3}$

III. $-20 \leq 2 - 3x \Rightarrow x \leq \frac{22}{3}$

$2 - 3x \leq 36 \Rightarrow -34 \leq 3x \Rightarrow x \geq \frac{-34}{3}$

Q.17 Answer:

70

Solution:

Let x be the marks obtained by student in the annual examination. Then, $\frac{62+48+x}{3} \geq 60$

or $110 + x \geq 180$

or $x \geq 70$

Thus, the student must obtain a minimum of 70 marks to get an average of at least 60 marks.

Q.18 Answer:

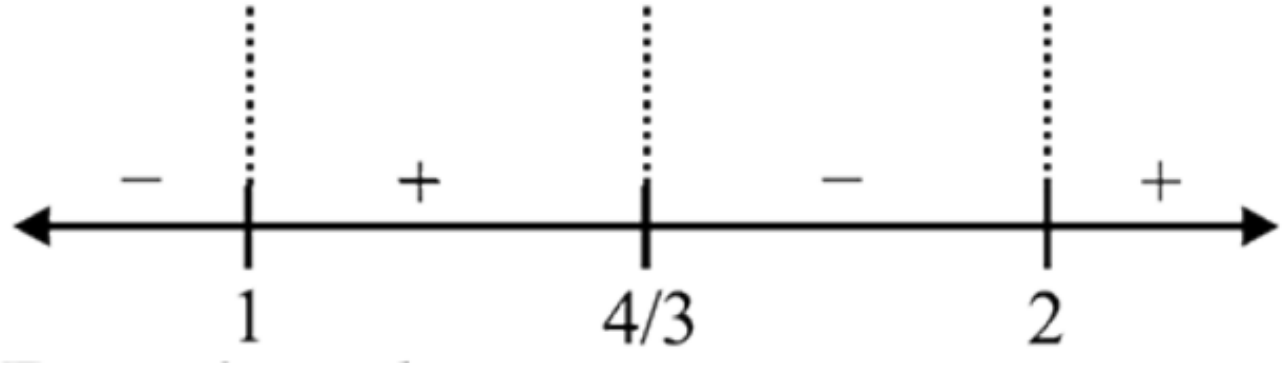
$(0, \log_2(4/3)) \cup (1, \infty)$

Solution:

@IITJEE_Advanced

Put $2^x = t$. Then $t > 0$. Now, given inequality becomes $\frac{1}{t-1} > \frac{2}{2-t} \quad \Rightarrow \frac{1}{t-1} - \frac{2}{2-t} > 0 \Rightarrow \frac{2-t-2t+2}{(t-1)(2-t)} > 0$

$\Rightarrow \frac{4-3t}{(t-1)(2-t)} > 0 \Rightarrow \frac{\left(t-\frac{4}{3}\right)}{(t-1)(t-2)} > 0$



Irom sign scheme we get $1 < t < 4/3$ or $t > 2$

$\Rightarrow \quad 1 < 2^x < 4/3$ or $2^x > 2$

$\Rightarrow x \in (0, \log_2(4/3)) \cup (1, \infty)$

Q.19 Answer:

$x + \log_2(1 - 2^x)$

Solution:

$2^{2y} - 2^y + 2^x(1 - 2^x) = 0$

put $2^y = \mathrm{t}$

$t^2 - t + 2^x(1 - 2^x) = 0$ where $t_1 = 2^{y_1}$ and $t_2 = 2^{y_2}$

$t_1 t_2 = 2^x(1 - 2^x)$

$2^{y_1 + y_2} = 2^x(1 - 2^x)$

$y_1 + y_2 = x + \log_2(1 - 2^x)$

Q.20 Answer:

more than 2000

Solution:

Given, $\mathrm{C(x)} = 26000 + 30\mathrm{x}$ and $\mathrm{R(x)} = 43\mathrm{x}$

$\therefore \quad$ Profit $= \mathrm{R(x)} - \mathrm{C(x)} = 43\mathrm{x} - (26000 + 30\mathrm{x})$

$= 13x - 26000$

For some profit, $13x - 26000 > 0\ 13x > 26000 \quad \Rightarrow x > 2000$

@IITJEE_Advanced